

Figure 6.3 Fuel filter



Figure 6.4 Air filter

and creates problem in reciprocating movement of the piston up and down. The incomplete combustion causes formation of pollutants in the exhaust. Hence fuel filter is used in path way of the fuel supply line to avoid the above said problems. It removes the dust particles and sends the purified fuel to the fuel injector and to the combustion chamber. Figure 6.3 shows the Fuel Filter

6.4. AIR FILTER

In any diesel or petrol engine for achieving combustion of the fuel sufficient amount of oxygen or pure air is required. Hence before sending the atmospheric air into the carburettor in petrol engines or directly into the combustion chamber in diesel engines, the air must be purified. In automobiles there are number of designs in air filters based on the engine used. If it is a gasoline engine air filter is mounted at the carburettor inlet, where as the filter is fitted at the intake manifold in diesel engine. There are different types air filters used in vehicles. Figure 6.4 shows the Air Filter. They are

1. Dry Type Air Cleaner (filter)
2. Oil bath type Air Cleaner (filter)
3. Oil coated air filter (Oil wetted type air cleaner)

Air filters applications (Uses of Air cleaner)

Though there are many usages of the air filters in many different applications, the important usages of the air filter in automobile are,

1. To send clean air without dust and contaminates to the carburettor.
2. To avoid mal-functioning of the engine and prevent damages of the engine parts due to incomplete combustion
3. To reduce the pressure difference occurs in the intake manifold and lowers the noise level. Figure 6.4 shows the Air Filter

6.4.1 Dry Type Air Cleaner

In this type of air filters, the cover, filtering part and bottle housing are coupled with each other. When the air is inducted to the filter through the filter cover, the dust particles and contaminations in the air are removed. Then purified air passes

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